Part I

Homological algebra

Chapter 1

Abelian categories

1.1 Additive categories

Definition 1.1.1 (Additive Category)**.** A Category A is **additive** if it verifies:

A.1) There exists a Zero object $0 \in \mathsf{A}$.

A.2) A has finite Products.

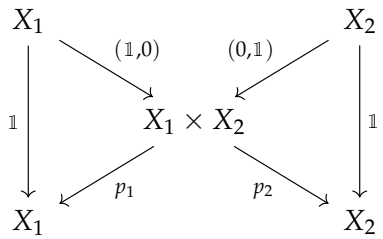
A.3) For every pair of objects $X, Y \in \mathsf{A}$ the set $\mathrm{Hom}_{\mathsf{A}}(X, Y)$ is endowed with an abelian group structure and the composition maps are bilinear with respect to the group operation.

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Lemma 1.1.1. *Finite Products and Coproducts coincide in an Additive Category.*

Proof. Consider $X_1, X_2 \in \mathsf{A}$. We only need to check that $X_1 \times X_2 \cong X_1 \amalg X_2$ since the case for a finite collection of elements follows directly by induction. To do this, we will see that $X_1 \times X_2$ satisfies the universal property of the Coproduct.

Consider the following diagram



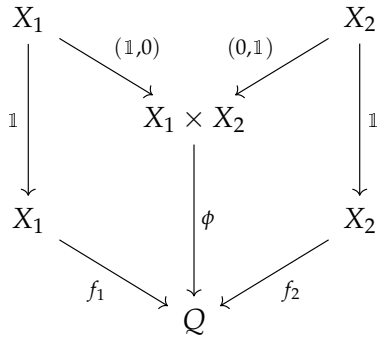
Note that for any pair of morphisms $(\alpha_1, \beta_1), (\alpha_2, \beta_2) : R \to X_1 \times X_2$ we have that

$$(\alpha_1, \beta_1) + (\alpha_2, \beta_2) = (\alpha_1 + \alpha_2, \beta_1, \beta_2)$$

Therefore if p_1, p_2 are two projections on $X_1 \times X_2$ whose sum is the identity we have that

$$(\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2 = \mathbb{1}_{X_1 \times X_2}$$

Consider now two morphisms $f_i : X_i \to Q$. Then its sum $\phi = f_1 \circ p_1 + f_2 \circ p_2$ produces the following diagram



Furthermore, this is unique. Indeed, consider $\phi' : X_1 \times X_2 \to Q$ another morphism filling the diagram. Then

$$\begin{aligned}
\phi &= \phi \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi \circ (id, 0) \circ p_1 + \phi \circ (0, \mathbb{1}) \circ p_2 \\
&= f_1 \circ p_1 + f_2 \circ p_2 \\
&= \phi' \circ (\mathbb{1}, 0) + \phi' \circ (0, \mathbb{1}) \\
&= \phi' \circ ((\mathbb{1}, 0) \circ p_1 + (0, \mathbb{1}) \circ p_2) \\
&= \phi' \circ \mathbb{1}_{X_1 \times X_2} \\
&= \phi'
\end{aligned}$$

Therefore $X_1 \times X_2$ verifies the universal property of $X_1 \amalg X_2$. □

Notation. If $\{X_i\}_{i=1}^n$ is a finite set of objects in an Additive Category we will denote its Product (and therefore by Lemma 1.1.1 its Coproduct) by $\bigoplus_{i=1}^n X_i$.

Lemma 1.1.2. *A morphism $\varphi : A \to B$ in an Additive Category is a monomorphism if and only if for every morphism $\xi : Z \to A$*

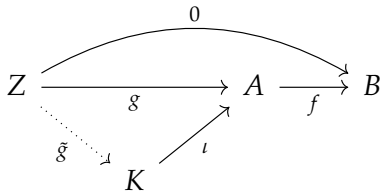
$$\varphi \circ \xi = 0 \implies \xi = 0$$

Dually φ is an epimorphism if and only if for every morphism $\xi : B \to Z$

$$\xi \circ \varphi = 0 \implies \xi = 0$$

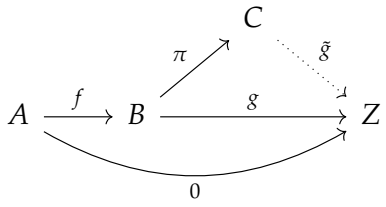
Definition 1.1.2 (Additive Functor)**.** Let A, B be two Additive Categorys. A Functor $F : \mathsf{A} \to \mathsf{B}$ is **additive** if it preserves the abelian group structure of all Hom-sets. ⌟

Definition 1.1.3 (Kernel)**.** Let $f : A \to B$ be a morphism in an Additive Category A. A morphism $\iota : K \to A$ is a **kernel** of f if $f \circ \iota = 0$ and for every morphism $g : Z \to A$ such that $f \circ g = 0$ there exists a unique $\tilde{g} : Z \to K$ making the following diagram commute



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Definition 1.1.4 (Cokernel)**.** Let $f : A \to B$ be a morphism in an Additive Category A. A morphism $\pi : B \to C$ is a **cokernel** of f if $\pi \circ f = 0$ and for every morphism $g : B \to Z$ such that $\pi \circ g = 0$ there exists a unique $\tilde{g} : C \to Z$ making the following diagram commute

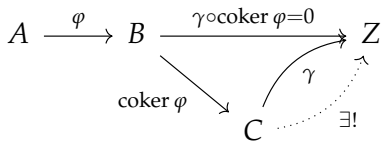


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Lemma 1.1.3. *In an Additive Category Kernels are monomorphisms and Cokernels are epimorphisms.*

Proof. We will only show that Cokernels are epimorphisms. The statement about Kernels follows analogously.

Let $\varphi : A \to B$ be a morphism and let $\operatorname{coker} \varphi : B \to C$ be its Cokernel. Consider another morphism $\gamma : C \to Z$ such that $\gamma \circ \operatorname{coker} \varphi = 0$. Since $(\gamma \circ \operatorname{coker} \varphi) \circ \varphi = 0$ by definition of the Cokernel we have



Therefore since $\gamma \circ \operatorname{coker} \varphi = 0 = 0 \circ \operatorname{coker} \varphi$ uniqueness forces $\gamma = 0$ and thus by Lemma 1.1.2 φ is an epimorphism. □

Lemma 1.1.4. *Let $\varphi : A \to B$ be a morphism in an Additive Category. Then φ is a monomorphism if and only if $0 \to A$ is its Kernel and φ is an epimorphism if and only if $B \to 0$ is its Cokernel.*

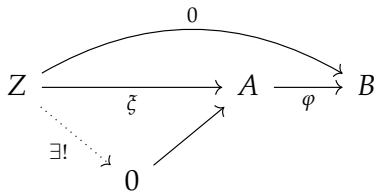
Proof. We will do the Kernel one.

$\Longrightarrow$ Suppose φ is a monomorphism.

If $\xi : Z \to A$ is a morphism such that $\varphi \circ \xi = 0$ then by Lemma 1.1.2 $\xi = 0$. In particular ξ factors uniquely through $0 \to A$, i.e: $0 \to A$ is the Kernel of φ.

$\Longleftarrow$ Suppose now that $0 \to A$ is the Kernel of φ.

Consider a morphism $\xi : Z \to A$ such that $\varphi \circ \xi = 0$. By the Kernel properties we have the following diagram



Therefore by uniqueness $\xi = 0$, making φ a monomorphism by Lemma 1.1.2.

□

1.2 Abelian categories

Definition 1.2.1 (Abelian Category)**.** An Additive Category A is **abelian** if it verifies

A.4) Kernels and Cokernels exist in A.

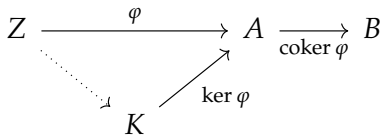
Furthermore every monomorphism is the Kernel of some morphism and every epimorphism is the Cokernel of some morphism.

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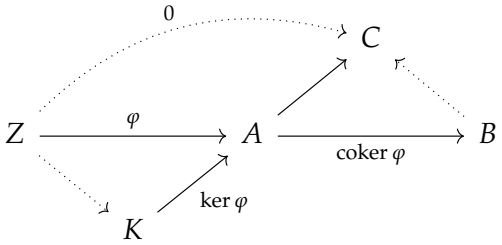
Lemma 1.2.1. *In an Abelian Category A every Kernel is the Kernel of its Cokernel and every Cokernel is the Cokernel of its Kernel.*

Proof. Lets check Cokernels this time.

Let coker $\varphi : A \to B$ be the Cokernel of a morphism $\varphi : Z \to A$. Since A is abelian φ has a kernel $\ker \varphi : K \to A$. Note that $Z \to A \to B$ is 0, therefore by the Kernel universal property we have



Consider a morphism $A \to C$ such that $(K \to A \to C) = 0$. Now using the Cokernel universal property we obtain

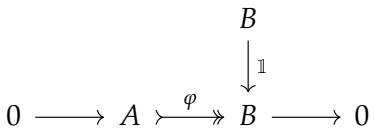


Therefore coker φ satisfies the universal property of the Cokernel of $\ker \varphi : K \to A$. □

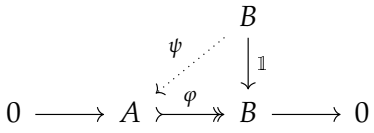
Lemma 1.2.2. *Let $\varphi : A \to B$ be a morphism in an Abelian Category* A. *If φ is both a monomorphism and epimorphism then φ is an Isomorphism.*

Proof. By Lemma 1.1.4 the Kernel of φ is $0 \to A$ and the Cokernel is $B \to 0$. Furthermore by Lemma 1.2.1 φ is the Cokernel of $0 \to A$ and the Kernel of $B \to 0$.

Consider the following diagram

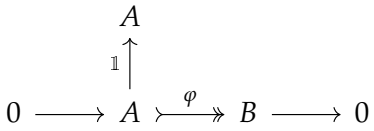


Since $(B \to B \to 0) = 0$ and φ is the Kernel of $B \to 0$ we have

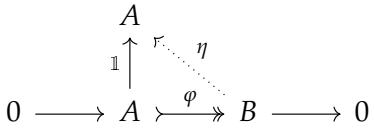


Therefore since $\varphi\psi = \mathbb{1}$ we have shown that φ has a right-inverse.

Consider now the diagram



Since $(0 \to A \to A) = 0$ and φ is the Cokernel of $0 \to A$ we have



Therefore since $\eta\varphi = \mathbb{1}$ we have shown that φ has a left-inverse. □

Theorem 1.2.1 (Freyd-Mitchell theorem)**.** *Let* A *be a small Abelian Category. Then there exists a fully faithfull* `def:exact-functor` $\mathsf{A} \to R\text{-}\mathsf{Mod}$ *for a suitable (possibly non-commutative) ring* R*, where* R-Mod *is the category of left* R*-modules.*

Chapter 2

Old

2.1 Complexes and (co)homology

From now on, we will be working in a fixed Abelian Category A. By the Freyd-Mitchell theorem we can work with objects and morphisms using set-theoretic tools when needed.

Definition 2.1.1 (Complex)**.** A sequence of objects and morphisms in an Additive Category A

$$\cdots \xrightarrow{d_{i-2}} C_{i-1} \xrightarrow{d_{i-1}} C_i \xrightarrow{d_i} C_{i+1} \xrightarrow{d_{i+1}} \cdots$$

is a **chain complex** if for every $i \in \mathbb{N} \quad d_i \circ d_{i-1} = 0$. We denote a chain complex as $(C_\bullet, d_\bullet)$ or simply as $C_\bullet$.

By reversing the order, we say that a sequence

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is a **cochain complex** if for every $i \in \mathbb{N} \quad d_i \circ d_{i+1} = 0$. We denote a cochain complex as $(C^\bullet, d^\bullet)$ or simply as $C^\bullet$. ⌟

Remark 2.1.1. In purelly algebraic terms, chains are cochains are *equivalent*. We will adopt the cochain notation. ▽

Definition 2.1.2 (Exact complex)**.** A Complex in an Abelian Category

$$\cdots \xrightarrow{d^{i+2}} C^{i+1} \xrightarrow{d^{i+1}} C^i \xrightarrow{d^i} C^{i-1} \xrightarrow{d^{i-1}} \cdots$$

is **exact at** C^i (or simply at position i) if

$$\operatorname{coker} d^{i+1} \circ \ker d^i = 0$$

A complex is **exact** if it is exact at every position. ⌟

Definition 2.1.3 (Image and coimage)**.** Let $\varphi : A \to B$ be a morphism in an Abelian Category.

The **image** of φ is $\operatorname{im}\varphi = \ker(\operatorname{coker}\varphi)$ and the **coimage** of φ is $\operatorname{coim}\varphi = \ker(\operatorname{coker}\varphi)$.

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Definition 2.1.4. Let $C^\bullet$ be a Complex in an Abelian Category A. We define the ***i*-th cohomology group** of $C^\bullet$ as

$$\mathrm{H}^i(C^\bullet) = {}^{\ker d^i}\!/_{\operatorname{im} d^{i-1}}$$

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Definition 2.1.5 (Category of complexes)**.** Let A be an Abelian Category. The **Category of Complexes** $\mathsf{C}(\mathsf{A})$ consists of:

1. Cochain complexes in A as objects.
2. For $(M^\bullet, d^\bullet_{M^\bullet}), (N^\bullet, d^\bullet_{N^\bullet}) \in \mathsf{C}(\mathsf{A})$, $\mathrm{Hom}_{\mathsf{C}(\mathsf{A})}(M^\bullet, N^\bullet)$ consists of the commutative diagrams of the form

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} & \xrightarrow{d^{i+1}_{M^\bullet}} & \cdots \\
 & & \downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^i} & & \downarrow{\scriptstyle \alpha^{i+1}} & & \\
\cdots & \longrightarrow & N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1} & \xrightarrow{d^{i+1}_{N^\bullet}} & \cdots
\end{array}$$

   We denote by $\alpha^\bullet : M^\bullet \to N^\bullet$ the morphism determined by the collection of α^i.

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Lemma 2.1.1. *The Category of complexes of an Abelian Category is an Abelian Category.*

Lemma 2.1.2. *Let A be a Abelian Category and let $\mathsf{C}(\mathsf{A})$ be its Category of complexes.*

Then for every integer i, the assignement $\mathrm{H}^i : M^\bullet \mapsto H^i(M^\bullet)$ taking a Complex $M^\bullet$ to its i-th is a Functor $\mathsf{C}(\mathsf{A}) \to \mathsf{A}$.

Proof. We want to show that for all Complexes $M^\bullet, N^\bullet \in \mathsf{C}(\mathsf{A})$ and $i \in \mathbb{Z}$, H^i induces in a natural way a homomorphism of abelian groups

$$\mathrm{Hom}_{\mathsf{C}(\mathsf{A})}(M^\bullet, N^\bullet) \to \mathrm{Hom}_{\mathsf{A}}(\mathrm{H}^i(M^\bullet), \mathrm{H}^i(N^\bullet))$$

Let $\alpha^\bullet : M^\bullet \to N^\bullet$ be a morphism in $\mathsf{C}(\mathsf{A})$. Let's consider the relevant of the commutative diagram defining $\alpha^\bullet$:

$$\begin{array}{ccccc}
M^{i-1} & \xrightarrow{d^{i-1}_{M^\bullet}} & M^i & \xrightarrow{d^i_{M^\bullet}} & M^{i+1} \\
\downarrow{\scriptstyle \alpha^{i-1}} & & \downarrow{\scriptstyle \alpha^i} & & \downarrow{\scriptstyle \alpha^{i+1}} \\
N^{i-1} & \xrightarrow{d^{i-1}_{N^\bullet}} & N^i & \xrightarrow{d^i_{N^\bullet}} & N^{i+1}
\end{array}$$

By the commutativity of the right square we have that $d^i_{N^\bullet} \circ \alpha^i = \alpha^{i+1} \circ d^i_{M^\bullet}$ is 0 on $\ker d^i$. Therefore the restriction of α^i to $\ker d^i_{M^\bullet}$ factors through $\ker d^i_{N^\bullet}$. Composing with the projection we get a morphism

$$\ker d^i_{M^\bullet} \to \mathrm{H}^i(N^\bullet)$$

Since this morphism is 0 when restricted to $\operatorname{im} d^{i-1}_{M^\bullet}$ by the commutativity of the left square we get that α^i induces a morphism

$$\mathrm{H}^i(M^\bullet) \to \mathrm{H}^i(N^\bullet)$$

in A as we wanted. □

Remark 2.1.2. We can view as a Functor $\mathsf{C}(\mathsf{A}) \to \mathsf{C}(\mathsf{A})$ by placing each object $\mathrm{H}^i(M^\bullet)$ in degree i and connecting them using 0 morphisms, obtaining a complex $\mathrm{H}^\bullet(M^\bullet)$.

Note that the of this particular Complex verifies

$$\mathrm{H}^\bullet(\mathrm{H}^\bullet(M^\bullet)) = \mathrm{H}^\bullet(M^\bullet)$$

▽